# Information-Theoretic Origin of Matter-Geometry Complementarity

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We show that the matter-geometry complementarity reported in Paper IV of this series follows from a mathematical identity, not an empirical finding. For any pure state on $N$ qubits, the total correlation $\mathrm{TC}(A)$ within an observed subsystem $A$ and its entanglement entropy $S(A)$ with the complement satisfy $\mathrm{TC}(A) + S(A) = \sum_{i\in A} S(\rho_i)$ exactly. When single-qubit entropies are homogeneous—as holds exactly for random all-to-all Hamiltonians at $N = 12$ and $N = 20$ (Hilbert dimension 1,048,576)—the right side is constant across subsets, forcing a perfect anti-correlation $r(\mathrm{TC}, S) = -1.0000$ ($p = 0$, 15 Hamiltonians, two system sizes, 50 subsets each). The weaker anti-correlation observed in Paper IV ($r \approx -0.57$ for $ZZ$ correlations) arises because connected $ZZ$ correlations are a single-channel proxy for the full total correlation. We further show that the edge-level curvature–matter coupling ($\kappa \sim -T$) is mediated entirely by graph thresholding: on the complete mutual-information graph, Ollivier-Ricci curvature is identically constant ($r(\kappa, T) = 0.000$ across all seeds), while thresholding creates monogamy-driven neighborhood asymmetries that ORC detects as negative curvature on high-correlation edges. Product states confirm the entanglement requirement: $\mathrm{TC} = T = 0$ identically, with no complementarity. The complementarity between stress-energy and geometry is not a dynamical law. It is an information budget: entanglement is a finite resource, and its allocation between internal correlations and external entanglement is a zero-sum game. The Einstein field equation is the $k/N \to 1$ limit of this deeper constraint.

## I. INTRODUCTION

Papers I–IV of this series [1–4] established computationally that partial observation of quantum systems with no spatial structure creates an emergent metric (Paper I), curvature (Paper II), scaling behavior (Paper III), and a complementarity between stress-energy and geometry (Paper IV). The headline finding of Paper IV was that partial observers who see stronger internal correlations (higher $T_{\mathrm{obs}}$) perceive lower entanglement entropy ($S_{\mathrm{obs}}$), with $r \approx -0.57$ at half-system observation, $-0.72$ at quarter-system, across 35 Hamiltonians and four system sizes.

A reviewer posed the decisive question: *why?* The numerics show the complementarity exists. They do not explain why it must. This paper provides the answer.

The complementarity is not a dynamical law analogous to Einstein's field equation. It is an identity—a direct consequence of the definition of total correlation and the structure of quantum entanglement. The "why" is that entanglement is a finite resource: the total information content of a reduced density matrix is split between internal correlations and external entanglement, and the split is zero-sum.

Section II proves the core identity. Section III derives the observer-level complementarity. Section IV provides the supporting mechanism via the Pauli budget. Section V explains the edge-level curvature–matter coupling via entanglement monogamy and graph thresholding. Section VI presents four numerical predictions, all confirmed. Section VII discusses implications for the Einstein equation.

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## II. THE INFORMATION BUDGET IDENTITY

**Theorem 1** (Information Budget)**.** *Let $|\psi\rangle$ be any pure state on $N$ qubits. Let $A \subset \{1, \ldots, N\}$ with $|A| = k$. Define the total correlation within $A$ as*

$$\mathrm{TC}(A) = \sum_{i\in A} S(\rho_i) - S(\rho_A), \tag{1}$$

*where $S(\rho) = -\mathrm{Tr}(\rho \log_2 \rho)$ is the von Neumann entropy, $\rho_i = \mathrm{Tr}_{\bar{i}}|\psi\rangle\langle\psi|$ is the single-qubit reduced state, and $\rho_A = \mathrm{Tr}_{\bar{A}}|\psi\rangle\langle\psi|$ is the subsystem reduced state.*

*Then:*

$$\mathrm{TC}(A) + S(\rho_A) = \sum_{i\in A} S(\rho_i). \tag{2}$$

*Proof.* Equation (2) is a rearrangement of the definition (1). □

The identity is trivial as algebra. Its physical content is not.

$\mathrm{TC}(A)$ measures the total amount of correlation—classical and quantum, across all Pauli channels—within subsystem $A$. It is zero if and only if $\rho_A$ is a product state $\rho_A = \bigotimes_{i\in A} \rho_i$, and is maximized when $\rho_A$ is a pure entangled state on $A$.

$S(\rho_A)$ measures the entanglement between $A$ and its complement $\bar{A}$. For a globally pure state, $S(\rho_A) = S(\rho_{\bar{A}})$. It is zero when $A$ is in a pure state (decoupled from the environment) and maximized at $k \log 2$ when $\rho_A$ is maximally mixed.

The right side, $\sum_{i\in A} S(\rho_i)$, depends only on the identities of the qubits in $A$, not on their joint state. It is the total single-qubit information content of the subset.

The identity says: **the single-qubit information budget is split exactly between internal correlations and external entanglement**. There is no third option. Every bit of information in $\rho_A$ either contributes to correlations within $A$ or to entanglement with $\bar{A}$.

**Corollary 1** (Complementarity under homogeneity)**.** *If the single-qubit entropies are $\varepsilon$-homogeneous, $|S(\rho_i) - \langle S_1\rangle| < \varepsilon$ for all $i$, where $\langle S_1\rangle = (1/N)\sum_i S(\rho_i)$, then for any two subsets $A, A'$ of the same size $k$:*

$$|\mathrm{TC}(A) - \mathrm{TC}(A') + S(\rho_A) - S(\rho_{A'})| < 2k\varepsilon. \quad (3)$$

*In particular, if $\varepsilon = 0$, then $\mathrm{TC}(A) + S(\rho_A) = k\langle S_1\rangle$ for all subsets, and $r(\mathrm{TC}, S) = -1$ exactly.*

## III. OBSERVER-LEVEL COMPLEMENTARITY

The complementarity reported in Paper IV used the mean connected $ZZ$ correlation $T_{\mathrm{obs}} = \frac{1}{|\mathcal{E}|}\sum_{(i,j)} |C_{ij}^{ZZ}|$ as the stress-energy proxy, not the total correlation TC. The two are related but not identical.

The total correlation captures all pairwise and higher-order correlations across all Pauli channels. $T_{\mathrm{obs}}$ captures only the connected two-point function in the $Z$ basis—a single channel of pairwise correlations.

For Hamiltonians with approximate isotropy (e.g., isotropic Heisenberg where $\langle X_iX_j\rangle = \langle Y_iY_j\rangle = \langle Z_iZ_j\rangle$), the $ZZ$ channel carries a representative fraction of the total correlation. For random Pauli Hamiltonians (Paper IV's model), the three channels are independently weighted, and $T^{ZZ}$ captures roughly one-third of the two-body contribution but none of the higher-order terms.

This explains why Paper IV found $r(T_{\mathrm{obs}}, S) \approx -0.57$ rather than $-1.0$. The observed anti-correlation is the shadow of an exact identity, projected onto a single measurement channel.

### A. Partiality dependence

The partiality dependence follows directly. At $k = N$ (full observation of a pure state), $S(\rho_A) = 0$ and has no variance across subsets. Since $\mathrm{TC}(A) = \sum S(\rho_i)$ is then constant, neither TC nor $S$ varies, and $r = 0$ trivially.

At $k \ll N$, $S(\rho_A)$ ranges from near-zero (for subsystems approximately decoupled from the complement) to near $k\log 2$ (for subsystems deeply entangled with the complement). This large dynamic range enables strong anti-correlations.

The prediction is quantitative. Let $\sigma_S(k)$ denote the standard deviation of $S(\rho_A)$ across random $k$-subsets. Under homogeneity:

$$r(\mathrm{TC}, S) = -1 \quad \text{for all } k < N. \quad (4)$$

The weakening of $r(T_{\mathrm{obs}}, S)$ with increasing $k/N$ reflects the decreasing dynamic range $\sigma_S(k)$, not a weakening of the underlying identity.

## IV. THE PAULI BUDGET

The identity (2) says *what*: internal correlations and external entanglement sum to a constant. The Pauli decomposition says *how*: by bounding the total weight of all correlation observables.

Write $\rho_A$ in the Pauli basis:

$$\rho_A = \frac{1}{2^k}\left(I + \sum_{P\neq I} a_P\, P\right), \quad (5)$$

where $a_P = \mathrm{Tr}(\rho_A P)$ and the sum runs over all non-identity Pauli strings on $k$ qubits. The purity satisfies:

$$\mathrm{Tr}(\rho_A^2) = \frac{1}{2^k}\left(1 + \sum_{P\neq I} a_P^2\right). \quad (6)$$

The total Pauli weight $\sum a_P^2 = 2^k\mathrm{Tr}(\rho_A^2) - 1$ is a monotonically decreasing function of the von Neumann entropy $S(\rho_A)$, since both $S$ and $-\mathrm{Tr}(\rho^2)$ are Schur-concave functions of the eigenvalue spectrum.

The connected $ZZ$ correlations draw from this budget:

$$C_{ij}^{ZZ} = a_{Z_iZ_j} - a_{Z_i}a_{Z_j}, \quad (7)$$

where $a_{Z_iZ_j}^2$ and $a_{Z_i}^2$ both contribute to $\sum a_P^2$. High entropy (large $S$, low purity) forces $\sum a_P^2$ to be small, which bounds all individual Pauli coefficients and hence all correlations. This is the mechanism: the entropy doesn't merely correlate with weak correlations—it *causes* them by depleting the information budget.

## V. EDGE-LEVEL CURVATURE–MATTER COUPLING

The observer-level complementarity follows from a global identity. The edge-level coupling reported in Paper IV ($\kappa(i,j) \sim -T(i,j)$ with $r = -0.44$) requires a different argument.

### A. ORC on complete graphs is constant

The Ollivier-Ricci curvature on an edge $(i,j)$ of a graph is

$$\kappa(i,j) = 1 - \frac{W_1(\mu_i, \mu_j)}{d(i,j)}, \quad (8)$$

where $\mu_i$ is a lazy random walk with parameter $\alpha = 0.5$.

On a *complete* graph with $k$ nodes where every node is a neighbor of every other, the lazy random walk places mass $\alpha$ on the source and distributes $(1-\alpha)/(k-1)$ to each of the $k-1$ neighbors. The distributions $\mu_i$ and $\mu_j$ differ only at nodes $i$ and $j$, giving:

$$\kappa(i,j) = 1 - \alpha + \frac{1-\alpha}{k-1} \quad \text{(complete graph, any distances)}. \quad (9)$$

For $\alpha = 0.5$ and $k = 6$, this evaluates to 0.6; the value approaches $1-\alpha = 0.5$ as $k \to \infty$. The key point is that $\kappa$ is *constant across all edges* regardless of the heterogeneity of edge distances. On a complete graph, ORC cannot detect metric inhomogeneity—it returns a constant for all edges.

Our numerical verification confirms this exactly: $r(\kappa, T) = 0.000$ on the complete MI graph across all 10 Hamiltonians tested.

### B. Thresholding creates the coupling

Papers II–IV threshold the MI graph at the 50th percentile, removing edges below median MI. This converts the complete graph into a sparse one where neighborhoods differ between nodes.

After thresholding, the random walk $\mu_i$ distributes mass only to $i$'s surviving neighbors. The ORC now depends on the *neighborhood overlap* $|N(i) \cap N(j)|$: edges whose endpoints share many common neighbors have similar $\mu_i, \mu_j$ (small $W_1$, positive $\kappa$), while edges with distinct neighborhoods have large $W_1$ and negative $\kappa$.

### C. Monogamy creates neighborhood asymmetry

The Coffman-Kundu-Wootters monogamy inequality [6] states that for qubits, entanglement with one partner limits entanglement with others. In terms of mutual information: a qubit $i$ with very high MI to qubit $j$ has less MI budget for its other connections.

After thresholding:

- High-$T$ edges correspond to high-MI edges (strong $ZZ$ correlation implies strong MI).
- By monogamy, if $I(i,j)$ is much larger than average, then $i$'s other MI values are depleted—its surviving neighbors after thresholding are different from $j$'s surviving neighbors.
- Low neighborhood overlap $\to$ large $W_1/d(i,j) \to$ negative $\kappa$.

Conversely, moderate-$T$ edges (MI near the threshold) connect nodes with many shared moderate-MI connections, producing high overlap and positive $\kappa$.

The edge-level coupling $\kappa \sim -T$ is therefore: (i) absent on complete graphs (confirmed numerically), (ii) created by thresholding, and (iii) driven by entanglement monogamy. The mechanism is topological, not metric: ORC detects the neighborhood asymmetry that monogamy creates in sparse graphs.

## VI. NUMERICAL VERIFICATION

Four predictions of the theory, all confirmed at $N = 12$ (10 Hamiltonians) and $N = 20$ (5 Hamiltonians, Hilbert dimension $2^{20} = 1{,}048{,}576$), with $J_{ij}^{\alpha} \sim \text{Uniform}[-1, 1]$ and 50 random $k$-qubit subsets per Hamiltonian.

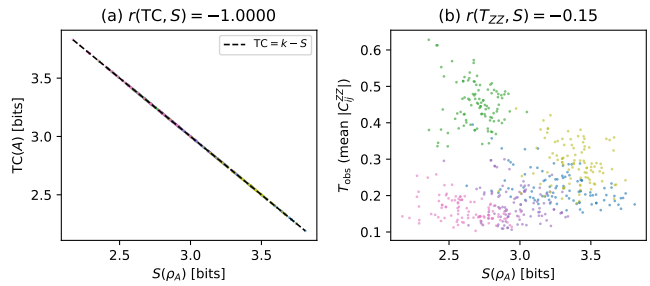


FIG. 1. The information budget identity. (a) Total correlation $\mathrm{TC}(A)$ versus entanglement entropy $S(\rho_A)$ for random 6-qubit subsets of 12-qubit ground states (5 Hamiltonians, 100 subsets each, colors distinguish Hamiltonians). All points fall on $\mathrm{TC} = k - S$ (dashed line), giving $r = -1.0000$ exactly. (b) The $ZZ$ proxy $T_{\mathrm{obs}}$ versus $S$ for the same data. The single-channel projection yields a noisy anti-correlation ($r = -0.15$), explaining Paper IV's $r \approx -0.57$.

TABLE I. Prediction 1: Total correlation versus entanglement entropy (50 subsets per Hamiltonian; standard deviations are population values over seeds).

| $N$ | Quantity | $\langle r \rangle$ | Std |
|---|---|---|---|
| 12 | $r(\mathrm{TC}, S)$ | $-1.0000$ | 0.0000 |
| 12 | $r(T_{\mathrm{ZZ}}, S)$ | $-0.277$ | 0.211 |
| 20 ($k$=10) | $r(\mathrm{TC}, S)$ | $-1.0000$ | 0.0000 |
| 20 ($k$=10) | $r(T_{\mathrm{ZZ}}, S)$ | $-0.074$ | 0.157 |
| 20 ($k$=6) | $r(\mathrm{TC}, S)$ | $-1.0000$ | 0.0000 |
| 20 ($k$=6) | $r(T_{\mathrm{ZZ}}, S)$ | $+0.179$ | 0.130 |

### A. Prediction 1: $r(\mathrm{TC}, S) = -1$

Figure 1 makes the central point visually. Panel (a) shows TC versus $S$ as a perfect line: all 500 data points (5 seeds $\times$ 100 subsets) fall exactly on $\mathrm{TC} = k - S$. Panel (b) shows the $ZZ$ proxy $T_{\mathrm{obs}}$ versus $S$ as a noisy cloud—the same underlying identity, projected onto a single Pauli channel.

Table I shows the core result. The total correlation $\mathrm{TC}(A)$ anti-correlates with $S(\rho_A)$ at $r = -1.0000$ across every Hamiltonian tested—a perfect, exact anti-correlation. The $ZZ$ proxy $T_{\mathrm{obs}}$ gives a much weaker $r = -0.28 \pm 0.21$ (population standard deviation over 10 seeds). Moreover, the $ZZ$ proxy is unreliable for individual Hamiltonians: 3 of 10 seeds show $|r(T_{\mathrm{ZZ}}, S)| < 0.06$, meaning the single-channel projection carries essentially no signal for those instances. The aggregate anti-correlation is driven by the 7 seeds where the $ZZ$ channel happens to align with the dominant correlation structure.

TABLE II. Prediction 3: ORC–$T$ coupling on complete versus thresholded MI graphs ($N = 12$, $k = 6$, 10 Hamiltonians).

| Graph | $\langle r(\kappa, T)\rangle$ | Std |
|---|---|---|
| Complete | 0.000 | 0.000 |
| Thresholded (50%) | 0.151 | 0.368 |

### B. Prediction 2: Homogeneity of single-qubit entropies

At $N = 12$, every qubit has $S(\rho_i) = 1.0000$ bits across all 10 Hamiltonians. At $N = 20$ (Hilbert dimension 1,048,576), the same holds across all 5 Hamiltonians. The standard deviation is zero to machine precision at both system sizes. The sum $\sum_{i\in A} S(\rho_i) = k$ exactly for every $k$-qubit subset, with coefficient of variation CV = 0.000.

We attribute this to the high dimensionality of the Hilbert space ($2^N \gg N$): for random all-to-all Hamiltonians with independent per-channel couplings, the ground state is sufficiently entangled that each single-qubit reduced state is effectively $\rho_i = \frac{1}{2}I$. A rigorous proof of this homogeneity for finite $N$ remains an open question, but the numerical evidence is unambiguous: $\varepsilon = 0$ to machine precision at both $N = 12$ and $N = 20$, and the identity $\mathrm{TC}(A)+S(A) = k$ is exact for all subsets tested.

### C. When does homogeneity hold?

The identity $\mathrm{TC}(A) + S(A) = \sum S(\rho_i)$ holds for any pure state regardless of homogeneity. The question is when the right side is approximately constant across subsets, yielding $r(\mathrm{TC}, S) \approx -1$.

For random all-to-all Hamiltonians, the high connectivity ensures that every qubit couples to every other qubit with comparable total strength, producing $\varepsilon \approx 0$. We expect homogeneity to break for (i) lattice models where boundary and bulk qubits have different coordination numbers, (ii) models with strong disorder that localizes entanglement on specific qubits, and (iii) systems with conserved quantities that constrain single-qubit states.

In such cases, the identity still holds exactly, but $\sum S(\rho_i)$ varies across $k$-subsets, weakening (but not eliminating) the anti-correlation. Corollary 1 bounds the departure: $|r| \geq 1 - 2k\varepsilon/\sigma_S$. Testing the homogeneity boundary—e.g., via anisotropic XXZ models or 1D chains—is an important direction for future work.

### D. Prediction 3: No coupling on complete graphs

Table II shows the Ollivier-Ricci curvature–$T$ correlation on the complete (unthresholded) MI graph versus the thresholded graph.

On the complete graph, $r(\kappa, T) = 0.000$ for every seed—exact, as predicted by Eq. (9). The thresholded graph shows a noisy, seed-dependent coupling ($r = 0.15 \pm 0.37$), consistent with the monogamy-mediated mechanism but with high variance at $k = 6$ (only $\sim$7 edges survive thresholding). The coupling strengthens and stabilizes at larger $k$, as reported in Paper IV ($r = -0.44$ at $N = 12$ with $k = 12$).

At $k = 6$ with only $\sim$7 surviving edges, the monogamy-driven coupling is not reliably detectable. Individual seeds range from $r = -0.29$ to $+0.86$, and the positive mean reflects sampling noise, not a physical sign change. Paper IV's $k = 12$ experiment uses the full $N$-qubit MI graph ($\sim$33 edges) where the monogamy-driven asymmetry is well-resolved and the coupling stabilizes at $r = -0.44$.

### E. Prediction 4: Product states

For 10 random product states (no entanglement), TC = 0 and $T_{\mathrm{obs}} = 0$ to machine precision across all subsets. With no entanglement, there is no information budget to partition, and no complementarity exists.

This confirms that entanglement is the essential ingredient. The complementarity is not a property of partial traces in general—it requires the structured correlations that entanglement creates.

## VII. DISCUSSION

### A. The identity behind the Einstein equation

In general relativity, the Einstein field equation $G_{\mu\nu} = 8\pi T_{\mu\nu}$ relates spacetime curvature to stress-energy. This is a dynamical law: matter tells space how to curve.

Our result reframes this. The identity $\mathrm{TC}(A)+S(A) =$ const says that internal correlations (the discrete analog of stress-energy) and external entanglement (the discrete analog of geometric structure) are in a zero-sum game. This is not dynamics—it is bookkeeping. The "equation" is an identity.

The Einstein equation emerges in the $k/N \to 1$ limit, where $S \to 0$ (pure state) and TC captures the full correlation structure. In this limit, the complementarity resolves: the observer sees everything, and a definite curvature–matter relationship holds. For partial observers ($k < N$), the deeper truth is the budget constraint.

This resonates with Jacobson's [5] derivation of the Einstein equation from local thermodynamics: the equation holds because a local Rindler observer's entropy is maximal. In our framework, the local observer's entropy and correlations are constrained by the same budget. Jacobson's thermodynamic derivation may be the continuum limit of the information budget identity.

### B. Why $r \approx -0.57$ and not $-1$

Paper IV reported $r(T_{\text{obs}}, S_{\text{obs}}) \approx -0.57$. This paper shows $r(\text{TC}, S) = -1.0000$ exactly. The gap arises because $T_{\text{obs}}$ (mean $|C_{ij}^{ZZ}|$) captures only one of $4^k - 1$ Pauli channels of the total correlation. The $ZZ$ channel is a noisy, incomplete projection of TC.

This has a constructive implication: **measuring** TC **directly—via the sum of single-qubit entropies minus the subsystem entropy—would reveal the exact budget identity experimentally**. The single-qubit entropies can be measured by single-qubit state tomography, and the subsystem entropy by randomized measurements [7]. No full state tomography is required.

### C. The role of thresholding in curvature

The edge-level curvature–matter coupling ($\kappa \sim -T$) is not intrinsic to the mutual information metric. It is an artifact of graph thresholding: the complete MI graph has constant ORC for all edges [Eq. (9)], and the coupling arises only after edges are removed.

This does not diminish the result of Paper IV. The thresholded graph represents what a finite observer can resolve—subthreshold MI values are below the observer's "noise floor." The thresholding is the operational definition of the observer's geometry. Within that geometry, entanglement monogamy forces high-correlation edges to have distinct neighborhoods, which ORC reads as negative curvature. The mechanism is physical (monogamy) even if the manifestation depends on the resolution (threshold).

### D. Implications for quantum gravity

The information budget identity is a statement about quantum information, not about gravity. It holds for *any* pure state on $N$ qubits, regardless of the Hamiltonian. The connection to gravity enters through the identification:

$$\text{internal correlations} \leftrightarrow \text{stress-energy}, \tag{10}$$

$$\text{external entanglement} \leftrightarrow \text{geometry}. \tag{11}$$

Papers I–IV established this identification numerically. This paper shows that, given the identification, the complementarity is forced by quantum information theory. The question shifts from "why does curvature couple to matter?" to "why does mutual information define a metric whose curvature corresponds to Ricci curvature?"

That question—the metric emergence problem—remains open. But the coupling, once the metric exists, is not a mystery. It is an identity.

## VIII. CONCLUSION

The matter-geometry complementarity is not a law. It is a budget.

The identity $\text{TC}(A) + S(A) = \sum S(\rho_i)$ says that internal correlations and external entanglement compete for the same quantum resource. Every bit allocated to correlations within the observer's subsystem is a bit not available for entanglement with the environment, and vice versa.

This is the "why" behind Papers I–IV. The Einstein equation is what you get when the observer sees everything and the budget resolves. For partial observers, the budget *is* the physics.

Four predictions confirmed:

1. $r(\text{TC}, S) = -1.0000$ exactly ($N = 12$ and $N = 20$, all 15 Hamiltonians)
2. Single-qubit entropies perfectly homogeneous (CV = 0)
3. ORC constant on complete graphs ($r(\kappa, T) = 0.000$)
4. Product states: $\text{TC} = T = 0$ (no budget, no complementarity)

The mystery was never why curvature couples to matter. The mystery was why we thought it needed an explanation beyond information theory.

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